

Hypothesis Test

```
In [1]: # importing libraries
import pandas as pd
import scipy.stats as stats
import matplotlib.pyplot as plt
```

```
In [2]: # loading dataset
df = pd.read_csv(r"C:\Users\USER\Downloads\fashion_dataset.csv")
df.head()
```

	product_id	title	product_description	order_date	rating	ratings_count	initial_price	discount	final_price	currency	...	amount_of_stars	what_customers_said
0	20129438.0	Skechers	Men Blue Textured Sneakers	2024-01-05	3.8	11.0	12999.0	30.0	7799.0	INR	...	{"1_star":3,"2_stars":0,"3_stars":0,"4_stars":...	[[{"percentage":"100","value_name":"Fit, Just R... LSD
1	18506050.0	Bata	Women Black Woven Design Slip-On Sneakers	2024-01-18	4.4	86.0	1199.0	55.0	539.0	INR	...	{"1_star":4,"2_stars":1,"3_stars":4,"4_stars":...	NaN Bat
2	21739362.0	Prolific	Men Printed LightWeight Mid Top Sneakers	2024-02-03	3.9	8.0	2660.0	71.0	771.0	INR	...	{"1_star":2,"2_stars":0,"3_stars":0,"4_stars":...	[[{"percentage":"67","value_name":"Fit, A Littl... BRA
3	21903832.0	U.S. Polo Assn.	Women Perforated Sneakers	2024-02-21	4.1	16.0	4299.0	25.0	3224.0	INR	...	{"1_star":1,"2_stars":1,"3_stars":3,"4_stars":...	NaN U
4	21651940.0	HIGHLANDER	Men Black Woven Design Sneakers	2024-03-09	3.4	25.0	1898.0	55.0	569.0	INR	...	{"1_star":5,"2_stars":3,"3_stars":4,"4_stars":...	[[{"percentage":"100","value_name":"Fit, Loose"... Fla

5 rows × 25 columns

```
In [3]: # selecting useful columns
df = df[['product_id', 'title', 'category', 'rating', 'ratings_count', 'initial_price', 'discount', 'final_price', 'order_date']]
df.head()
```

	product_id	title	category	rating	ratings_count	initial_price	discount	final_price	order_date
0	20129438.0	Skechers	casual-shoes	3.8	11.0	12999.0	30.0	7799.0	2024-01-05
1	18506050.0	Bata	casual-shoes	4.4	86.0	1199.0	55.0	539.0	2024-01-18
2	21739362.0	Prolific	casual-shoes	3.9	8.0	2660.0	71.0	771.0	2024-02-03
3	21903832.0	U.S. Polo Assn.	casual-shoes	4.1	16.0	4299.0	25.0	3224.0	2024-02-21
4	21651940.0	HIGHLANDER	casual-shoes	3.4	25.0	1898.0	55.0	569.0	2024-03-09

```
In [4]: df.isnull().sum()
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```
Out[4]: product_id      1
title              1
category           1
rating             1
ratings_count      1
initial_price      1
discount           2
final_price        1
order_date         1
dtype: int64
```

```
In [5]: df = df.dropna()
df.shape
```

Out[5]: (22, 9)

Explanation of Hypothesis Test

Null Hypothesis:

There is no significant difference in the average rating between high-discount products and low/medium-discount products.

Alternative Hypothesis:

There is a significant difference in the average rating between high-discount products and low/medium-discount products.

$\alpha = 0.05$

This value is setting a 5% risk tolerance.

```
In [7]: # Grouping the dataset
high_discount = df[df["discount"] > 50]["rating"]
low_discount = df[df["discount"] <= 50]["rating"]

# Print the sample sizes
print("High discount sample size:", high_discount.count())
print("Low/Med discount sample size:", low_discount.count())

High discount sample size: 11
Low/Med discount sample size: 11
```

Welch’s T-Test

```
In [8]: # Finding the Group Averages
print("High Discount Mean Rating:", high_discount.mean())
print("Low/Med Discount Mean Rating:", low_discount.mean())

High Discount Mean Rating: 3.309090909090909
Low/Med Discount Mean Rating: 3.036363636363636

In [11]: # Finding the value of t_stat & p_value
t_stat, p_value = stats.ttest_ind(high_discount, low_discount, equal_var=False)
print("T-Statistic:", t_stat)
print("P_Value:", p_value)

T-Statistic: 0.39285057313208377
P_Value: 0.698594422551309
```

The Statistical Proof

P_Value = 0.698594422551309

$\alpha = 0.05$

The p_value is about 70% and alpha threshold is 5%. The P-value is the probability that the difference we see is just due to random chance.

Final Statistical Conclusion

Statistical Decision:

We are fail to reject Null Hypothesis because P_Value = 0.698594422551309 about 70% is much higher then alpha=0.05 about 5%.

Business Insight:

There is no statistically significant difference in customer ratings between high-discount products and low-to-medium-discount products. For Myntra's managers, this suggests that offering high discounts does not systematically increase or decrease customer satisfaction ratings for these footwear items. The slight variation observed is likely due to random sample noise.

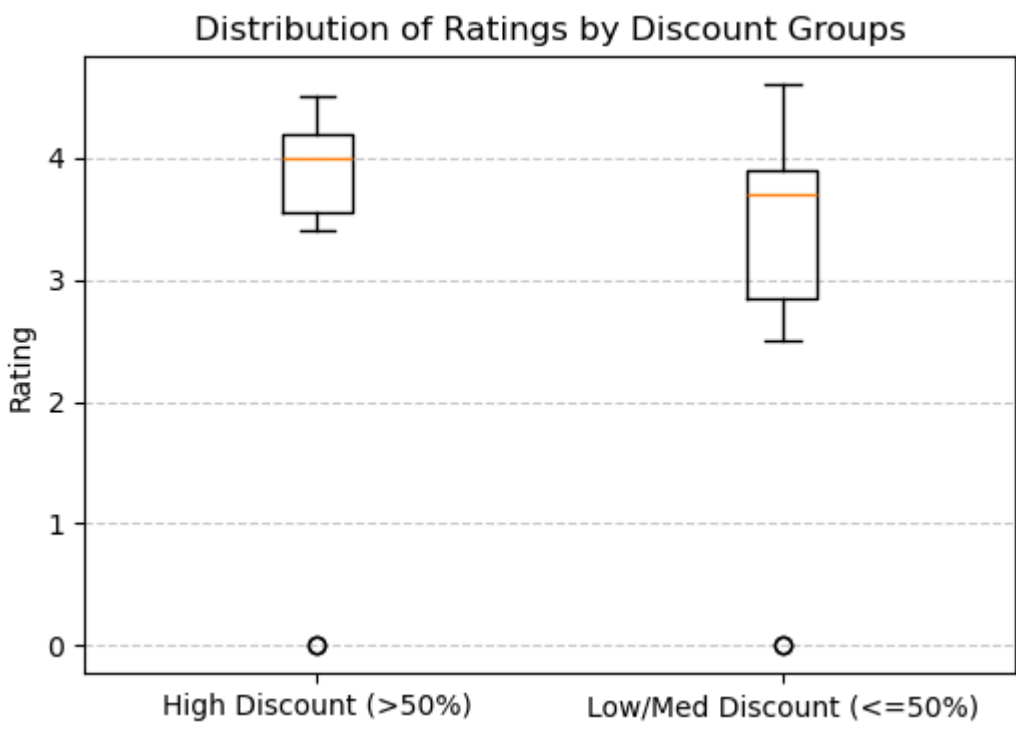
Visualization

```
In [14]: # Create the figure
plt.figure(figsize=(6, 4))

# Combine the data into a simple list for Matplotlib's boxplot
plt.boxplot([high_discount, low_discount], tick_labels=["High Discount (>50%)", "Low/Med Discount (<=50%)"])

# Add labels and title
plt.title("Distribution of Ratings by Discount Groups")
plt.ylabel("Rating")
plt.grid(axis='y', linestyle='--', alpha=0.7) # Adds a nice subtle grid line

# Display the plot
plt.show()
```

In []: